OOI KAI HONG

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PROFESSIONAL PROFILE

Interdisciplinary biomedical researcher with experience in cell biology, advanced microscopy, quantitative bioimage analysis, clinical research, trial documentation, research ethics and scientific writing. Directed scientific requirements and iterative review for OrganelleScope, a desktop microscopy workbench implemented through Codex that combines earlier colocalization and line-profiling tools, and contributed to an object-detection workflow for microtubule analysis. Currently pursuing a Master of Artificial Intelligence at Monash University, with developing Python and AI/ML skills supported by structured study and practical projects.

CORE COMPETENCIES

Biomedical Research	Cell Biology | Molecular Biology | Cell Culture | Biochemistry | Immunostaining | Western Blot | Co-IP | Molecular Cloning | PCR
Microscopy & Bioimage	Confocal | Airyscan | STED | SIM | STORM | MINFLUX | Expansion Microscopy | Fiji/ImageJ | ZEN Blue | Quantitative Imaging
Clinical & Research	Clinical Trials | TMF/eTMF | Veeva Vault | Research Ethics | Good Clinical Practice | Systematic Review | Scoping Review | Qualitative Research | Scientific Writing
AI & Computing	Python (developing) | Machine Learning (project experience) | Object Detection (project experience) | Bioimage Analysis | Image Quantification | SQL (foundational) | GitHub | Scientific Software Prototyping

PROFESSIONAL EXPERIENCE

Research Admin | Faculty of Medicine, University of Malaya

02/2026 – 04/2026 | Kuala Lumpur, Malaysia

- Synthesized existing literature to identify and validate research instruments for a qualitative study focused on telenursing implementation.
- Co-authored a published scoping review on tele-nursing implementation, benefits, and barriers in postnatal and maternal-child healthcare.

Research Assistant | Hong Kong University of Science & Technology (Guangzhou Campus)

09/2023 – 11/2025 | Guangzhou, China

- Conducted biomedical research involving cell culture, biochemical assays, confocal microscopy, super-resolution microscopy, and quantitative image analysis.
- Investigated protein localization using confocal and super-resolution microscopy.
- Investigated the effects of kinase signalling proteins on microtubule nucleation.
- Contributed to development of automated bioimage quantification using object-detection machine learning.
- Assisted with manuscript review, editing, and preparation of scientific outputs.

Research Assistant | Faculty of Business & Economics, University of Malaya

05/2023 – 07/2023 | Kuala Lumpur, Malaysia

- Assisted with literature collection and survey-question finalization for research on financial initiatives supporting quality caregiving for older adults.
- Participated in participant recruitment and interview processes.
- Contributed to research resulting in publications in WoS-ESCI and Scopus-indexed journals.

Research Assistant | Faculty of Medicine, University of Malaya

12/2022 – 02/2023 | Kuala Lumpur, Malaysia

- Co-authored a systematic review on knee osteoarthritis interventions.

Trial Master File (TMF) Document Specialist 1 | Syneos Health Inc.

04/2021 – 10/2021 | Kuala Lumpur, Malaysia

- Conducted secondary quality control of Trial Master File documents using the Veeva Vault eTMF system.
- Met daily KPIs for document inspection and supported process efficiency.
- Collaborated with project teams to resolve document-quality issues.
- Maintained low/zero weekly error rates during document inspection.
- Received an Accountability Organization Badge.

Genetic Product Specialist Intern | Absolute Genetic Technologies

12/2020 – 02/2021 | Kuala Lumpur, Malaysia

- Interpreted genetic reports in English and Chinese.
- Assisted with survey data collection, analysis, and presentation.
- Created scientific and client-facing content and marketing materials using Canva.
- Collected biological samples to inform product development.

SELECTED SOFTWARE & COMPUTATIONAL PROJECTS

OrganelleScope

Desktop Microscopy Workbench | 2026

• Created OrganelleScope through Codex-led implementation, translating microscopy-analysis needs into software requirements. My role focused on defining scientific workflows, requesting features, reviewing application behaviour and providing iterative feedback. Codex generated and revised the code; I do not claim independent implementation of the underlying software.
• Specified workflows for calibrated 8/16-bit multichannel TIFF inspection, ROI measurements, line profiling, colocalization and 3D cuboid/proximity analysis within a combined workbench.
• Requested features for reviewed image batches, Excel/JSON exports, portable projects and analysis-review reports retaining source, settings and ROI provenance.
• Requested an optional local Gemma assistant for human-reviewed interpretation drafts, with source-support and coverage checks kept separate from quantitative measurements.

GitHub: github.com/mansonmyr/OrganelleScope

Probabilistic Colocalization Estimator (PCE) | OrganelleScope Predecessor

Scientific Software Prototype | 2026

- Directed requirements and review for the earlier ROI-centric tool for multichannel 8/16-bit immunofluorescence TIFF analysis; it subsequently became a predecessor and migration source for OrganelleScope.
- Brought correlation, thresholded overlap and regression-based readouts together with explicit preprocessing and exportable analysis records.
- AI coding tools generated and revised the implementation; my role focused on scientific requirements, feature requests, application review and feedback.

GitHub: github.com/mansonmyr/Probabilistic_Colocalization_Estimator

Quick Immunofluorescence Image Line Profiler (QILP) | OrganelleScope Source Tool

Scientific Software Prototype | 2026

- Directed requirements and review for a Codex-implemented browser-based profiler for 2–4 channel immunofluorescence TIFF images; selected profiling helpers, charting, TIFF export and tests were adapted into OrganelleScope.
- Supports calibrated line ROIs, background subtraction, peak detection, FWHM/AUC measurements, and CSV/JSON result export.
- Codex generated and revised the implementation; my role focused on scientific requirements, feature requests, application review and feedback.

GitHub: github.com/mansonmyr/Quick-IF-Line-Profiler

Automated Cytosolic Microtubule Detection Using Object Detection ML

Biomedical Computer Vision Prototype | 2026

- Prototyped an object-detection workflow for automated identification of cytosolic microtubule structures from fluorescence microscopy images.
- Workflow included microscopy image preparation, manual bounding-box annotation using LabelMe, dataset conversion, model training, model export, and interactive inference demonstration.
- AI-assisted implementation and prototyping; project experience rather than independent ML-engineering expertise.

GitHub: github.com/mansonmyr/Automated-cytosolic-microtubule-using-object-detection-ML

Semi-automated Quantification of In-vivo Microtubule Intensity Using Fiji/ImageJ Macros

Scientific Image Analysis Automation | 2026

- Created Fiji/ImageJ macros for semi-automated quantification of microtubule fluorescence intensity from immunofluorescence z-stack images.
- Workflow uses sum projection, filtering, Find Maxima detection, ROI generation, background correction, centrosomal intensity measurement, and spindle-pole analysis.
- Outputs quantitative measurements including centrosomal intensity, spindle-pole intensity, spindle length, and spindle microtubule intensity.

GitHub: github.com/mansonmyr/Semi-automated-quantification-of-in-vivo-microtubule-intensity-using-Fiji-ImageJ-macros

RESEARCH EXPERIENCE & PROJECTS

iENGAGE Summer Student Research Programme | Wellcome-Wolfson Institute for Experimental Medicine, Queen's University Belfast

07/2023 – 08/2023

- Completed desk-based research on B-cell immune response.
- Received training in data analysis, scientific writing, project design, and experimental techniques.
- Participated in journal clubs, seminars, and a research symposium.

Health Research Ethics Practicum / Summer Research Intern | University of Malaya

07/2022 – 09/2022 | Kuala Lumpur, Malaysia

- Contributed to mHealth research for HIV prevention in partnership with Yale University and the University of Connecticut.
- Conducted transcription and data analysis for focus groups and participated in community advisory board meetings.
- Contributed to a publication in JMIR Formative Research.

Master's Thesis | University of Malaya

2022 | Kuala Lumpur, Malaysia

- Research topic: Ethical Consideration on Clinical Trials in Low-and-Middle Income Countries: an Overview of Systematic Review.
- Presented research at the SIDCER-FERCAP International Conference (2022), and University of Malaya Medical Centre Research Carnival (2023).
- Published in Discover Global Society.

Bachelor's Bioinformatics Project | Sunway University

2020 | Kuala Lumpur, Malaysia

- Investigated G-quadruplex structures in HIV-1 CRF01_AE for potential antiviral development.
- Used QGRS Mapper and NCBI BLASTn to identify potential quadruplex-forming sequences across 1,335 strains.
- Investigated potential G-quadruplex folding across multiple HIV-1 genomic regions.
- Published a narrative review based on the project.

EDUCATION

Master of Artificial Intelligence | Monash University

08/2026 – Present

- Course: Introduction to Python Programming, Introduction to Computer Architecture, Mathematical Foundation for DS/AI, IT Research Method.

Master of Health Research Ethics | University of Malaya

10/2021 – 09/2022 | Kuala Lumpur, Malaysia

- CGPA: 3.61; collaborative degree programme with the Berman Institute of Bioethics, Johns Hopkins University.
- Full scholarship from the Fogarty International Centre of the United States National Institutes of Health (Grant R25TW010891).

Bachelor of Science (Hons) in Medical Biotechnology | Sunway University / Lancaster University, UK

03/2018 – 02/2021 | Malaysia

- Second Class Honours (Division One); CGPA: 3.38.

PUBLICATIONS

- Khati, A., Wickersham, J.A., Rosen, A.O., Luces, J.R.B., Copenhaver, N., Jeri-Wahrhaftig, A., Ab Halim, M.A., Azwa, I., Gautam, K., Ooi, K.H., & Shrestha, R. (2022). Ethical Issues in the Use of Smartphone Apps for HIV Prevention in Malaysia: Focus Group Study With Men Who Have Sex With Men. JMIR Formative Research, 6(12), e42939.
- Ooi, K.H., & Qi, Y. (2023). Unlocking the power of G-quadruplex for anti-HIV1 development. MEDS Clinical Medicine, 4(4), 49-52.
- Che Azmi, A., Ooi, K.H., Ab Aziz, N., Shahimi, S., & Lee, W.L. (2023). Financial Aspects of Elderly Care that Enhances their Quality of Life: A Scientometric Review. Journal of Scientometric Research, 12(3), 691-698.
- Che Azmi, A., Ooi, K.H., Shahimi, S., Ab. Aziz, N., & Silahudin, S. (2024). Financial Challenges in Music and Dance-based Interventions for Older Adults: Systematic Review. Malaysian Journal of Music, 13(1).
- Che Azmi, A., Ooi, K.H., Shahimi, S., & Ab. Aziz, N. (2025). Bibliometric analysis of the financial aspects of quality care for older people. Discover Global Society, 3(117).
- Han, L.J., Ooi, K.H., Krisna, V.B., & Phan, C.W. (2026). Ethical Challenges and Considerations in Decentralized Clinical Trial (DCTs): Insights from a Scoping Review. Therapeutic Innovation & Regulatory Science, 60, 904-924.
- Ooi, K.H., Ooi, P.B., & Phan, C.W. (2026). An umbrella review of ethical challenges in clinical trials conducted in low and middle income countries. Discover Global Society, 4(220).
- Ooi, K.H., Marina, J.R., Che, C.C., Nik Farid, N.D., & Abdullah, N.A. (2026). Implementation, Benefits, and Barriers of Tele-nursing in Postnatal and Maternal-Child Health Care: A Scoping Review. BMJ Innovations.
- Lee, J.SY., Gokilavanan, V., Ooi, K.H., & Tan, M.K.M. (2026). Beyond Lockdowns: Equity, Trust, and Social Justice in Malaysia's Pandemic Strategy. In: COVID-19 in the Pacific Rim and Beyond. Collaborative Bioethics, vol. 6. Springer, Cham.
- Lim, W.S., Ooi, K.H., Ooi, P.B., & Phan, C.W. (2026). Knowledge, Attitude, Awareness, Barriers, and Willingness to Participate in Clinical Trials: A Quantitative Survey among Undergraduate Students in Malaysia. Malaysian Journal of Medicine and Health Sciences. [Under review]

PROFESSIONAL DEVELOPMENT & AI LEARNING

- GCI World 2026 — Matsuo-Iwasawa Laboratory, The University of Tokyo | Participant | 09/2026 – Present
- Code in Place 2026 — Stanford University | 2026
- Python Basics for Data Science — IBM | Completed
- M1 in SQL — TalentLabs | Completed
- Higher Education Course in Artificial Intelligence — ENEB Business School | Completed
- Teaching English to Foreign Learner Certificate — Foreign Talent Research Centre, MOHRSS China | Completed
- Clinical Research Associate Certificate — CCRPS | Completed
- Social and Behavioral Responsible Conduct of Research — CITI Program | Completed
- Result-based Project Management — Inter-American Development Bank | Completed

INTERNATIONAL EDUCATION & LANGUAGE DEVELOPMENT

Hanyang International Summer School | Hanyang University

06/2026 – 07/2026 | Seoul, South Korea

- Completed beginner Korean language course.

TECHNICAL SKILLS

Laboratory: Cell Culture | Immunostaining | Crosslinking Co-IP | Western Blot | Sucrose Gradient Centrifugation | Molecular Cloning | PCR
Microscopy: LSM980 Confocal & Airyscan (Zeiss) | Stellaris 8 Confocal (Leica) | Axio Observer 7 | STED (Abberior STEDyCON) | SIM (Zeiss Elyra 7) | STORM (Abbelight) | MINFLUX (Abberior) | Expansion Microscopy
Flow Cytometry: BD FACSymphony A1 | FlowJo | Cell-cycle analysis
Bioimage Analysis: Fiji/ImageJ | ZEN Blue
Bioinformatics: GO & KEGG analysis | Clustal MSA | NCBI BLASTn | QGRS Mapper | TCGA
AI / Data: Python (developing) | Machine Learning (project experience) | Object Detection (project experience) | Automated Image Quantification | SQL (foundational) | GraphPad Prism | SPSS
Administrative / Digital: Veeva Vault | Tableau | Flourish | Canva | Microsoft Office | G Suite | Lucidchart | MindNode | Google Sites | Gantt Chart

LANGUAGES

- English — IELTS 7.0 (2026)
- Mandarin
- Malay
- Cantonese
- Korean — Beginner; Hanyang International Summer School (06/2026 – 07/2026)

CONFERENCE & SERVICE

Conference Support Volunteer | International Conference on Cell Organization & Genome Integrity 2024, HKUST(GZ)

12/2024 | Guangzhou, China

- Facilitated registration, session logistics, attendee support, and daily conference operations.
- Provided bilingual Chinese-English assistance to international delegates.
- Managed attendee flow to support smooth session transitions.

Oral Presenter | 22nd WHO SIDCER-FERCAP 2022, K-MEDI Hub

11/2022 | Daegu, Korea

- Presented topic on ethical consideration in clinical trial among low- and middle-income countries.